

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Botany**Paper: BOB-503 — Plant Ecology**

Time: Three hours

Max. Marks: 70

Note: Answer five questions in all, selecting two questions from Section-A and Section-B. Section-C is compulsory. The figures in the right-hand margin indicate marks.

SECTION – A

Question 1. Discuss the indices of alpha diversity and its components to measure species diversity. [15]

Question 2. Discuss various theories/models of mechanism of ecological succession. [15]

Question 3. Write in detail on major types of soil pollution and routes of land contamination. [15]

Question 4. Define teratogens. Briefly discuss factors affecting toxicity of chemical agents. [15]

SECTION – B

Question 5. Write descriptive notes on any two of the following: [7.5 × 2 = 15]

- (a) Population concepts
- (b) Global pattern of primary productivity
- (c) Energy flow

Question 6. Write critical notes on any two of the following: [7.5 × 2 = 15]

- (a) Global carbon cycle
- (b) Ecological niche
- (c) Importance Value Index (IVI)

Question 7. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Eutrophication
- (b) Thermal pollution
- (c) Greenhouse effect

Question 8. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Dose-response relationship
- (b) Concept of toxicity
- (c) Types of important toxicant groups

SECTION – C

Question 9. Answer the following in not more than 50 words each: [1 × 10 = 10]

- (a) Ecological pyramid
- (b) Sere
- (c) β -diversity
- (d) Humus
- (e) Nudation
- (f) “Silent Spring”
- (g) BOD
- (h) Photochemical smog
- (i) Name two chemicals that are toxic in their elemental form.
- (j) Stratification.

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